ID: 5252e606

The side length of a square is 55 centimeters (cm). What is the area, in cm^2, of the square?

A. 110

B. 220

C. $3{,}025$

D. $12{,}100$

ID: 59cb654c

The area of a square is 64 square inches. What is the side length, in inches, of this square?

A. 8

B. 16

C. 64

D. 128

ID: d0b6d927

A rectangle has an area of 63 square meters and a length of 9 meters. What is the width, in meters, of the rectangle?

A. 7

B. 54

C. 81

D. 567

ID: 0837c3b9

Triangle *ABC* and triangle *DEF* are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $DE = 2AB$ and the perimeter of triangle *ABC* is 20, what is the perimeter of triangle *DEF* ?

A. 10

B. 40

C. 80

D. 120

ID: c88183f7

A rectangle has a length of 13 and a width of 6. What is the perimeter of the rectangle?

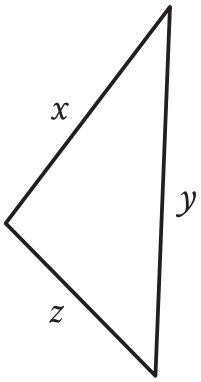
A. 12

B. 26

C. 38

D. 52

ID: 29e9b28c



Note: Figure not drawn to scale.

The triangle shown has a perimeter of 22 units. If $x = 9$ units and $y = 7$ units, what is the value of z, in units?

A. 6

B. 7

C. 9

D. 16

ID: 3453aafc

What is the area, in square centimeters, of a rectangle with a length of 36 centimeters and a width of 34 centimeters?

A. 70

B. 140

C. $1{,}156$

D. $1{,}224$

ID: f60bb551

The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

A. 9

B. 70

C. 315

D. 560

ID: 4420e500

What is the area of a rectangle with a length of 4 centimeters (cm) and a width of 2 cm?

A. $6\ \text{cm}^2$

B. $8\ \text{cm}^2$

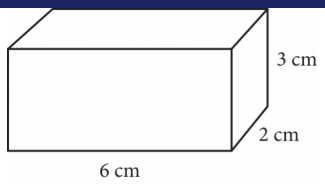
C. $12\ \text{cm}^2$

D. $36\ \text{cm}^2$

ID: 165c30c4

A rectangle has a length of 64 inches and a width of 32 inches. What is the area, in square inches, of the rectangle?

ID: d683a9cc



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh, where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?

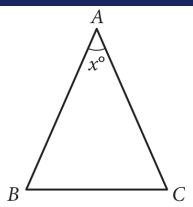
A. 36

B. 24

C. 12

D. 11

ID: c8d60e48



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of $67°$. What is the value of x?

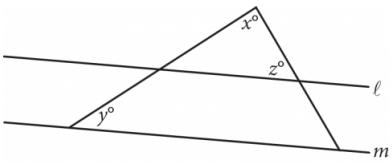
A. 36

B. 46

C. 58

D. 70

ID: a6dbad6b



Note: Figure not drawn to scale.

In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

A. 120

B. 100

C. 90

D. 80

ID: cbe8ca31

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98°. What is the measure of $\angle Z$?

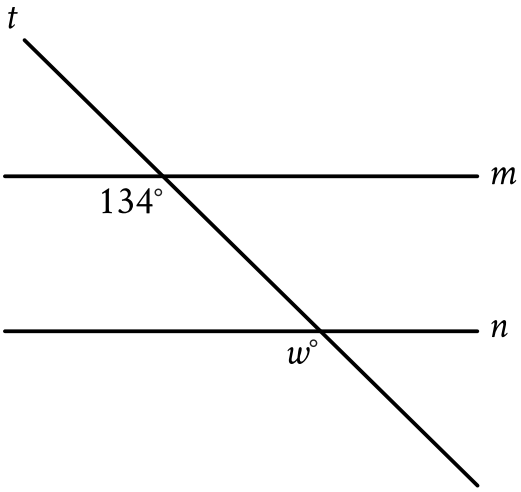
A. 58°

B. 74°

C. 122°

D. 212°

ID: c24e1bda



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n. What is the value of w?

A. **13**

B. **34**

C. **66**

D. **134**

ID: f9d40000

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66°. What is the measure of $\angle Z$?

A. 43°

B. 89°

C. 91°

D. 179°

ID: 3563d76d

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

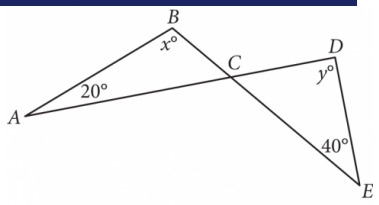
A. 10

B. 20

C. 30

D. 35

ID: dfc420b2



Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

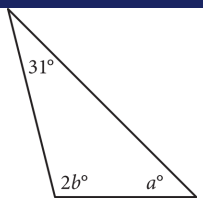
A. 100

B. 90

C. 80

D. 60

ID: 410bdbe6



In the triangle above, $a = 45$. What is the value of b ?

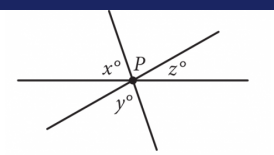
A. 52

B. 59

C. 76

D. 104

ID: 087cdcfd



Note: Figure not drawn to scale.

In the figure, three lines intersect at point *P*. If $x = 65$ and $y = 75$, what is the value of *z* ?

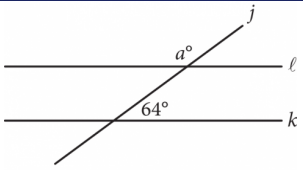
A. 140

B. 80

C. 40

D. 20

ID: 992f4e93



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel.

What is the value of a ?

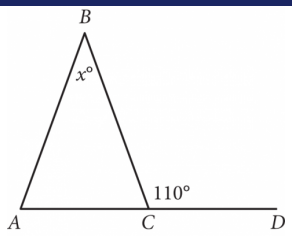
A. 26

B. 64

C. 116

D. 154

ID: 5733ce30



In the given figure, $\overline{AC}$ extends to point *D*. If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of *x* ?

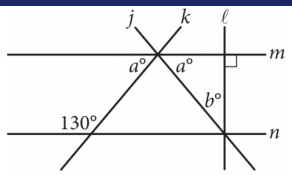
A. 110

B. 70

C. 55

D. 40

ID: 3828f53d



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel.
What is the value of b ?

A. 40

B. 50

C. 65

D. 80

ID: 42b4493b

In a right triangle, the measure of one of the acute angles is 51°. What is the measure, in degrees, of the other acute angle?

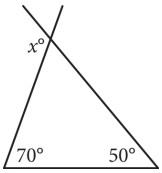
A. 6

B. 39

C. 49

D. 51

ID: 36200a38



In the figure above, two sides of a triangle are extended. What is the value of x ?

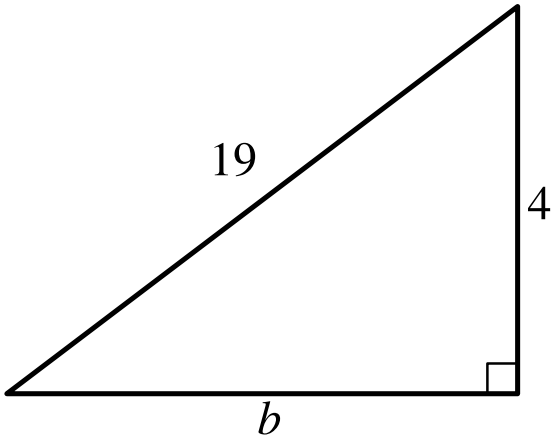
A. 110

B. 120

C. 130

D. 140

ID: b0c5ece5



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

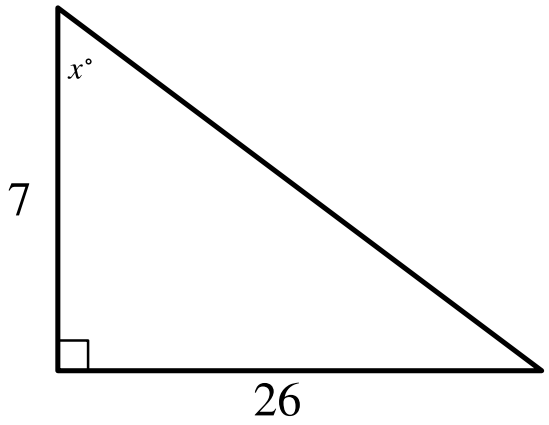
A. $4b = 19$

B. $4 + b = 19$

C. $4^2 + b^2 = 19^2$

D. $4^2 - b^2 = 19^2$

ID: 64c1f044



Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\tan x^\circ$?

A. $\frac{1}{26}$

B. $\frac{19}{26}$

C. $\frac{26}{7}$

D. $\frac{33}{7}$